🧠 MULTIPLE SCLEROSIS (MS)

Multiple Sclerosis is a chronic, progressive, autoimmune disorder that affects the **central nervous system (CNS)** — including the **brain and spinal cord**. It involves the **destruction of myelin**, the protective covering of nerves, which impairs communication between the brain and body.

📃 KEY TERMINOLOGY

- **Autoimmune** – (auto = self; immune = immune system) The immune system attacks the body's own cells.
- **Demyelination** – (de = removal; myelin = protective nerve sheath) Loss of myelin from nerve fibers.
- **Exacerbation** – (ex = out; acerbus = bitter) A flare-up or worsening of symptoms.
- **Remission** – (re = again; mittere = to send) Periods where symptoms improve or disappear.
- **Paresthesia** – (para = abnormal; esthesia = feeling) Tingling, numbness, or "pins and needles" sensation.

🧬 PATHOPHYSIOLOGY

- Immune system attacks **myelin sheath** → slows or blocks nerve impulses.
- Affects CNS only (brain and spinal cord; not peripheral nerves).
- Leads to **plaques** or scars on nerves → permanent damage over time.

🧠 TYPES OF MS

1. **Relapsing-Remitting (RRMS)** – most common; symptoms come and go.
2. **Primary Progressive** – steady worsening from onset.
3. **Secondary Progressive** – initially relapsing-remitting, then steady decline.
4. **Progressive-Relapsing** – rare; steady worsening with acute flare-ups.

⚠️ COMMON SIGNS & SYMPTOMS

- **Fatigue** (most common and debilitating)
- **Muscle weakness or spasticity** (stiff or tight muscles)
- **Vision changes** (blurry vision, diplopia = double vision, optic neuritis)
- **Numbness/tingling** (paresthesia)
- **Bladder and bowel dysfunction** (urgency, retention, constipation)
- **Ataxia** (a = without; taxia = coordination) – unsteady gait
- **Dysarthria** (slurred speech)
- **Dysphagia** (difficulty swallowing)
- **Cognitive changes** (memory loss, poor concentration)
- **Heat sensitivity** (Uhthoff's phenomenon – symptoms worsen with heat)

🔬 DIAGNOSIS

- **MRI** – shows brain and spinal cord lesions or plaques.
- **Lumbar puncture** – reveals ↑ **oligoclonal bands** in CSF (indicative of inflammation).
- **Evoked potentials** – tests slowed nerve responses.
- **Rule out** other neurological diseases.

💊 TREATMENT (NO CURE – SYMPTOM MANAGEMENT)

- Disease-Modifying Drugs (slow progression):
- **Interferon beta-1a/1b** – reduces relapses
- **Glatiramer acetate (Copaxone)** – mimics myelin
- **Natalizumab** – prevents immune cells from crossing blood-brain barrier
- **Fingolimod** – traps immune cells in lymph nodes

SYMPTOM MANAGEMENT:

- **Corticosteroids (methylprednisolone, prednisone)** – reduce inflammation during flares
- **Baclofen, tizanidine** – for muscle spasticity
- **Oxybutynin** – for bladder urgency
- **Bethanechol** – for urinary retention

- **Amantadine, modafinil** – for fatigue
- **Gabapentin, pregabalin** – for neuropathic pain

🧼 NCLEX NURSING PRIORITIES

- **Energy conservation** – schedule rest periods, avoid overheating
- **Fall prevention** – assistive devices, remove hazards
- **Bowel/bladder training** – timed voiding, fluid management
- **Vision care** – eye patch for diplopia, report vision changes
- **Monitor swallowing** – risk for aspiration with dysphagia
- **Teach** about avoiding triggers (heat, stress, illness, fatigue)
- **Immunosuppressive precautions** – infection prevention

🔍 NCLEX TEST STRATEGY TIPS

- Symptoms can vary **widely** and may come/go → tailor care to current condition
- During **flare-ups**, prioritize corticosteroids to reduce inflammation
- Encourage **cool environments** and AC (Uhthoff's phenomenon = heat worsens symptoms)
- MS affects CNS only → peripheral nervous system (PNS) is spared
- Be alert for **dysphagia, immobility, aspiration**, and fatigue-related complications

💡 NCLEX MEMORY AIDS

- **M** – Muscle weakness
- **S** – Sight issues (optic neuritis, diplopia)
- **FATIGUE** – #1 complaint
- **MS** – Multiple Symptoms that come and go